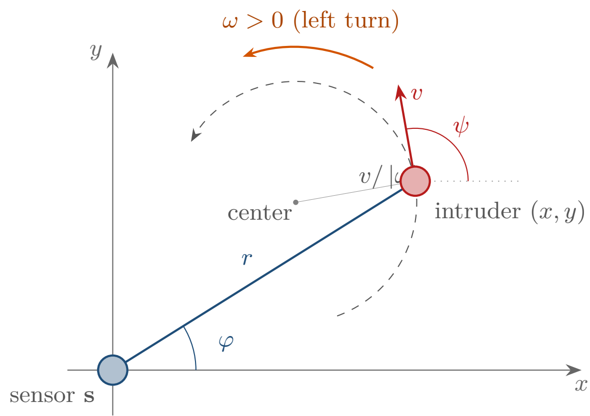

$$\text{state } \mathbf{x} = (x, y, v, \psi, \omega)^{\mathsf{T}}$$

$$\dot{x} = v\cos\psi,\ \dot{y} = v\sin\psi,\ \dot{\psi} = \omega$$

$$\text{path: arc of radius } v/\left|\omega\right|$$

$$\text{measurement } \mathbf{z} = (r, \varphi)^{\mathsf{T}}$$

$$r = \|(x, y) - \mathbf{s}\|$$

$$\varphi = \operatorname{atan2}(y - s_y,\ x - s_x)$$